MagTaxi: Induction Driven Airport Surface Mobility

Review #0

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Problem Statement & Urgency

- **Ground Fuel Waste:** Commercial jet engines are designed for high-altitude cruise, leading to severe thermodynamic inefficiency during low-speed ground taxiing[cite: 1, 2].
- **Breakaway Thrust Waste:** Heavy aircraft require high engine power simply to start moving in stop-and-go ground traffic[cite: 2].
- **Urgent Industry Need:**
  - *Net-Zero 2050 Goals:* Eliminating ground engine idling provides an immediate reduction in carbon emissions[cite: 2].
  - *Financial & Noise Restrictions:* High fuel consumption strains airline operating margins[cite: 2]; urban night-noise curfews restrict night flight windows[cite: 2].
  - *FOD Risk:* Low ground operation risks Foreign Object Debris (FOD) ingestion and costly engine damage[cite: 2].

Stakeholders & Target Audience

- **Primary Users:** Commercial airlines, flight crews, and airport ground control operators[cite: 2].
- **Secondary Stakeholders:** Civil aviation regulatory bodies (e.g., DGCA, FAA), airport developers, and aircraft OEMs[cite: 2].
- **Direct Beneficiaries:**
  - *Airlines & Crew:* Reduced fuel expense, lower engine overhaul frequency, and improved tarmac safety[cite: 2].
  - *Local Communities:* Elimination of ground emissions (CO_2, NO_x) and significant reduction in community noise[cite: 1, 2].

Proposed Solution: Project MagTaxi

- **Core System Concept:** Concrete-embedded Linear Induction Motor (LIM) tracks propel aircraft to and from gates without running main engines[cite: 1, 2].
- **Lightweight Nose Attachment:** An ultra-compact magnetic receiver (< 50 kg) with an emergency quick-release retrofitted onto the nose landing gear[cite: 2].
- **Energy Recovery (KERS):** Contactless magnetic braking captures deceleration energy and feeds power back into the airport storage grid[cite: 1, 2].
- **Target Improvements:** Up to a 90% reduction in ground fuel burn, zero localized taxi emissions, and complete elimination of taxi-phase FOD engine damage[cite: 2].

Comparison with Existing Alternatives

Method	Working Principle	Key Shortcomings
Jet Engines	Main engine thrust crawling[cite: 1]	High fuel burn, heavy emissions, FOD risk[cite: 2].
Tow-Tractors	External diesel/EV tugs (TaxiBot)[cite: 2]	Tarmac clutter, manual connection delay[cite: 2].
On-Board Motors	Gearbox-driven wheels (WheelTug)[cite: 2]	Heavy permanent weight penalty reduces payload/range[cite: 2].
MagTaxi	Ground-embedded LIM induction[cite: 1, 2]	**No flight weight penalty ($<$ 50 kg attachment)[cite: 2].**

Key Challenges & System Requirements

- **Engineering & Operational Challenges:**
  - *Cold Engine Warm-Up:* Coordinating precise engine start windows before entering the takeoff threshold[cite: 2].
  - *Interoperability:* Universal mounting interface compatible across different nose-gear geometries[cite: 2].
  - *Grid Surge Management:* Smart power management to handle sudden power demands during multiple aircraft movements[cite: 2].
- **Phased Rollout Strategy:**
  - *Phase 1 (Smaller Airstrips):* Initial validation on regional airfields to lower CapEx and test safety systems[cite: 2].
  - *Phase 2 (Major Hubs):* Installation along high-density bottleneck taxiways at major international hubs[cite: 1, 2].
  - *Phase 3 (Full Automation):* Seamless multi-terminal automated routing and smart-grid energy recycling[cite: 1, 2].

References



Team Aviators, *Project MagTaxi: Induction Driven Airport Surface Mobility Proposal*, Genesis Ideathon, Amrita Vishwa Vidyapeetham, 2026[cite: 1, 2].